

1st Test (sample)

1. Give both the adjacency list and the adjacency matrix representations of the following graph.
2. The transpose of a directed graph $G = (V, E)$ is the graph $G^T = (V, E^T)$ where $E^T = \{(v, u) \in V \times V \mid (u, v) \in E\}$. Thus G^T is G with all its edges reversed. Describe linear time algorithms (in the size of the input) for computing G^T from G for both *a*) the adjacency matrix and *b*) the adjacency list representations of G . Justify your answer.
3. We have three jugs of capacity 4, 7 and 10 liters. These are initially filled with ..., ..., and ... liters of water. In the goal state they should be filled with ..., ... and ... liters. What is the minimum number of steps of pouring water from one jug to another (until either one jug becomes empty or the other becomes full) needed to reach the goal state? Justify your answer.
4. Show how breadth-first search works on the following graph using vertex m as a source. Assume that each adjacency list is ordered alphabetically. Show the distance and parent values for each vertex, and show the queue at the beginning of each iteration of the while loop.
5. Show how depth-first search works on the following graph. Assume that the *for* loop of the main DFS procedure considers the vertices in alphabetical order, and assume that each adjacency list is ordered alphabetically. Show the discovery and finishing times for each vertex, and show a sample for each edge classification type (tree edge, forward edge, backward edge, cross edge).